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Extended depth of focus multiphoton microscopy via incoherent pulse splitting

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Abstract: We present a beam splitter mask that can be easily added to a multiphoton raster scanning microscope to extend the depth of focus five-fold at a small loss in lateral resolution. The method is designed for ultrafast laser pulses or other light-sources featuring a low coherence length. In contrast to other methods of focus extension, our approach uniquely combines low complexity, high light-throughput and multicolor capability. We characterize the point spread function in a two-photon microscope and demonstrate fluorescence imaging of GFP labeled neurons in fixed brain samples as imaged with conventional and extended depth of focus two-photon microscopy.

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1. Introduction

Imaging large volumes with raster scanning microscopes can be very time consuming as each voxel is acquired serially. This is particularly true if high numerical aperture (NA) objectives are employed that provide a high lateral resolution and a small depth of focus. To capture volumetric imaging data exceeding such a small depth, many planes have to be imaged by Z-stepping each focal plane, which limits the achievable imaging speed considerably. Instead, if axial image information can be sacrificed, it may be beneficial to extend the depth of focus such that a volume can be acquired in one lateral scan. In other words, volumetric information is projected into a single 2D image. This concept, referred to as extended depth of focus (EDF) imaging, is particularly attractive for sparsely populated structures that require high temporal resolution, such as functional imaging of neuronal activity [1,2].

Various beam shaping methods have successfully been employed for EDF imaging. These include different amplitude, phase, or combination filters in the pupil plane [3–5], or in some cases, specialized optical elements such as axicons [6,7]. Their implementation typically requires significant changes to a conventional microscope architecture [1]. Furthermore, if phase modulation is used, the associated phase masks or holograms are typically wavelength dependent, which prevents, or at least complicates, multi-color imaging.

A conceptually different approach of using an incoherent superposition as a means to extend the depth of focus was presented by Abrahamsson and Gustafsson [8]. Here, the back-pupil of the imaging objective was segmented into multiple annuli, which were made incoherent to one another. The exact relative phase between the different annuli did not matter, so long as the resulting beams were mutually incoherent. This concept increased the depth of focus of a widefield microscope by exploiting the short coherence length of fluorescence. However, we reasoned that this concept could also be applicable to beam shaping of ultrafast laser sources, which feature a similarly short coherence length. Thus, in this manuscript, we explore this

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concept and present numerical simulations and experimental measurements of two-photon EDF microscopy using an incoherent annular beam splitter mask. Importantly, we show that this mask can be easily introduced into existing two-photon laser scanning microscopes, and we highlight its potential for imaging applications in neuroscience by imaging GFP-labeled neurons in a fixed brain slice.

2. Method

2.1. Theory

The beam splitter mask, which is acting in a Fourier plane of the imaging system, consists of multiple concentric glass disks (Fig. 1(a)). An ultrafast laser pulse (typical pulse length in the tens of microns) is divided by the beam splitter mask into different annular beamlets (Fig. 1(b)), each of which is time-delayed and forms a focus in the front focal plane of the objective at slightly different arrival times. If the time delay introduced by each annulus largely exceeds the pulse duration of the laser, then the beamlets are mutually incoherent. If this condition is met, the resulting EDF focus is an incoherent superposition of all individual foci (Fig. 1(c)), which led us to term the technique incoherent pulse splitting. As shown in previous work using such a mask [8], for processes that scale linearly with intensity (e.g., one-photon excitation), sidelobe structures will surround the main lobe of the EDF focus. In our case, the non-linear nature of two-photon absorption suppresses the sidelobe structures and yields a smooth, gradual decay into the background (Fig. 7 in Appendix A).

As the overall point spread function (PSF) is the incoherent sum of the foci generated by each time-delayed beamlet, we must ensure that each beamlet produces a focus with the same extended depth-of-focus. Consider the diagram shown in Fig. 2(a). Here, we can see all the wavevectors admitted by the objective lens as a black line, with a subset highlighted in cyan. These wavevectors all lie on a spherical cap and have a magnitude of $k = n / \lambda_0$. Intensity variations within the focal region arise from the interference of different wavevectors [9]. As such, we can link the spatial extent of features laterally (in the XY plane) or axially (in the XZ plane) with the spread of wavevectors present along k_x and k_z.

Considering the lateral dimension first, we see from the diagram that the full spread of k_x components present within the spherical cap is given by $2k \sin \alpha_2$, leading to the well-known Abbe relation of $d_{\text{lateral}} = \lambda_0 / 2\,\text{NA}$ (where we have used the relation that $n \sin \alpha_2$ defines the NA). In the axial case, the spread of k_z wavevector components is given by $k (\cos \alpha_1 - \cos \alpha_2)$. In the case of using the full spherical cap, where $\alpha_1 = 0$, we obtain the relation $k (1 - \cos \alpha_2)$, giving

$$d_{axial} = \frac{\lambda_o}{n \times (1 - \cos \alpha_2)} \tag{1}$$

In Fig. 2(b), we see how the zones of the pupil mask map onto this spherical cap of wavevectors. For an aplanatic imaging system obeying the sine condition, such as commercially available microscope objective lenses, the flat entrance pupil of the objective (i.e. the one located near the shoulder) can be considered to orthographically project onto the spherical exit pupil. This allows us to map zones of the mask to sets of wavevectors for which we can use the geometric considerations detailed previously and shown in Fig. 2(a).

The central magenta zone maps to a small cap producing a standard focus, for which the axial extent is $\lambda_0 / (n \times (1 - \cos \alpha_2))$. For the other zones, α_1 becomes non-zero, so we must use the relation $\lambda_0 / (n \times (\cos \alpha_1 - \cos \alpha_2))$. This can be rewritten in terms of NA by using $\alpha = \sin^{-1}(NA/n)$ to produce a relation for the Bessel-like beams:

$$d_{axial} = \frac{\lambda_o}{n \times \left\{\cos\left[\sin^{-1}\left(\frac{NA_1}{n}\right)\right] - \cos\left[\sin^{-1}\left(\frac{NA_2}{n}\right)\right]\right\}} \tag{2}$$

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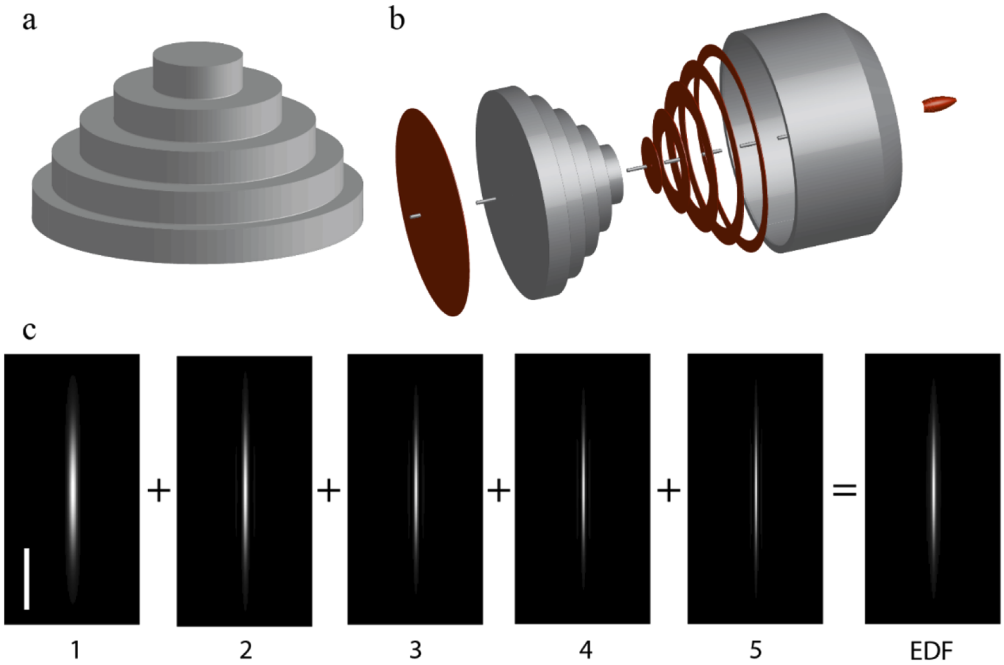

Fig. 1. Principle of extended depth of focus for ultrafast laser pulses. (a) Schematic of the beam splitter mask consisting of multiple glass disks of varying diameter. (b) Illustration of the working principle. An ultrafast laser pulse (red disk) is incident on the beam splitter mask. Portions of the pulse travelling through the different annular zones are time delayed into separate beamlets (red annuli). Each beamlet forms a focus at the front focal plane of the objective at a different time. (c) Numerical simulations of the squared intensity of the laser foci produced by the different beamlets. Numbers label annular zones, with 1 the innermost zone and 5 the outermost one. The effective EDF point spread function is the incoherent sum of all five foci. Scale bar: 10 um.

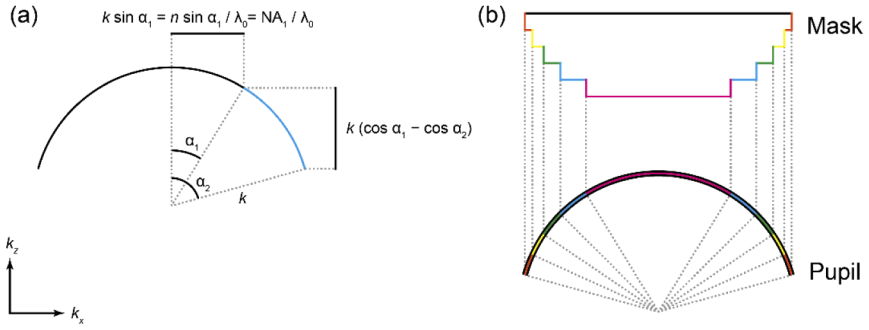

Fig. 2. Geometric considerations for calculating the depth-of-focus. (a) The Ewald sphere wavevector construction for a high-NA objective lens (NA_1), with a subset of wavevectors highlighted in cyan. α_1, α_2 angles defining the selected subset, k wavenumber, $\boldsymbol{\lambda_0}$ wavelength, k_X component of wavevector in x direction, k_Z component of wavevector in z direction (b) Mapping the pupil mask zones onto the Ewald sphere via the sine condition.

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Furthermore, the NAs can be converted to physical mask dimensions, as the back pupil diameter is equal to $2 \times f \times \mathrm{NA}$, where f is the back focal length of the objective. Thus, using these equations, we can use an iterative procedure to determine the required radii for a given depth-of-focus and number of annular zones. First, we calculate the central zone NA which produces a Gaussian-like PSF of the desired depth-of-field using Eq. (1). Next, we set the first annular zone to have an inner NA, $\mathrm{NA_{in}}$, equivalent to the NA just calculated for the Gaussian focus, and an outer NA, $\mathrm{NA_{out}}$, such that the Bessel-like focus has the same depth-of-field, as calculated by Eq. (2). This process is repeated for each additional annular zone using the previous outer NA as the inner NA until all annuli are considered or the maximum available NA of the lens is used (see Code 1 [11]). Using this procedure, we were able to confirm previous work that suggested the depth of focus scales linearly with the number of annular zones [8], and we mathematically prove this assertion in Appendix B.

2.2. Simulations

Numerical simulations to compute the electromagnetic field in the front focal plane of an objective were performed using the Debye theory [10], which is documented in the Code 1 [11]. In short, the simulation calculated the 3D electromagnetic field in the focal volume for each annular pupil zone individually. Of note, each pupil zone was assumed to be completely incoherent to the other zones, which allowed us to calculate each zone's contributions separately. Each electric field was converted to an intensity by squaring the modulus of the field, which was squared again to account for the non-linear intensity dependence of two-photon excitation. The resulting PSFs were then summed to form the overall EDF PSF. A conventional PSF, using the same maximal NA as in the EDF case, was also calculated to enable comparison.

As we wanted to match the physical parameters of the prefabricated mask used for our experimental results, we converted the annular diameter of the mask (1.8 mm, 2.5 mm, 3.1 mm, 3.6 mm, 4.0 mm) to NAs, taking into account the magnification between mask and pupil, and calculated the corresponding electric fields (see Code 1 [11]). We used a maximum NA of 0.67 by selecting the magnification of the pupil relay, which yielded in the simulation a lateral and axial resolution of 550 nm and 15.98 microns, as measured by their Full-Width Half-Maximum (FWHM). Compared to a conventional laser focus using an NA of 0.67, the EDF beam is 4.96-fold larger in the axial dimension (Fig. 3(b)). In contrast, the lateral FWHM of the EDF beam is only 7% larger than a conventional laser focus (Fig. 3(c)). Throughout this manuscript, we chose to characterize PSFs using the FWHM measure, as it is an established metric in other work on extended depth of focus [1] and hence allows comparisons of performance.

2.3. Experimental setup

Our setup was based on a homebuilt two-photon raster scanning microscope (Fig. 4). Ultrafast laser pulses (900nm wavelength, ~140fs pulse lengths) from a Ti:Sapphire oscillator (Chameleon Vision, Coherent Inc) passed through a Pockels cell (350-80-2, Conoptics) for power modulation and were expanded 6-fold by a Galilean telescope (AC254-50-B, AC254-300-B Thorlabs) and spatially cleaned up with a pinhole (PH-100, Newport). The expanded laser beam was incident on the beam splitter mask (part Nr. A12802-35-040 Hamamatsu Photonics). The mask was made of quartz glass and each layer had an approximate thickness of 400 microns. This resulted in a time delay of 0.72 ps between each zone at a wavelength of 900 nm, assuming a refractive index of 1.54 [12]. Therefore, the beam splitter mask fulfilled the requirements for incoherent superposition by exceeding the pulse duration (~140 fs) by a factor of five.

An iris was used to adjust the beam diameter to the size of the mask. To enable comparative measurements, the same iris was also used to create a low NA excitation beam in a configuration without the beam splitter mask (Fig. 4). The beam splitter mask was conjugated by a telescope onto the first galvo mirror (6215H, Cambridge technology). The galvo mirror was conjugated

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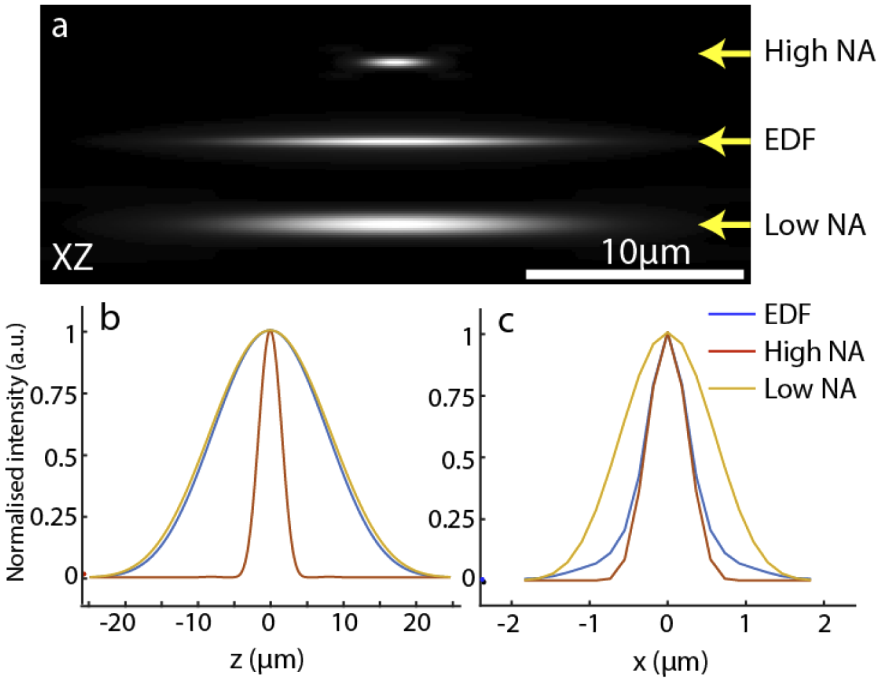

Fig. 3. Numerical simulation of PSFs. (a), from top to bottom: high NA PSF using an NA of 0.67 and 900 nm wavelength. Middle: EDF PSF. Bottom: low NA PSF to match axial extent of the EDF PSF. The low NA corresponds to 0.3. (b-c) Axial and lateral cross-sections through the PSFs shown above.

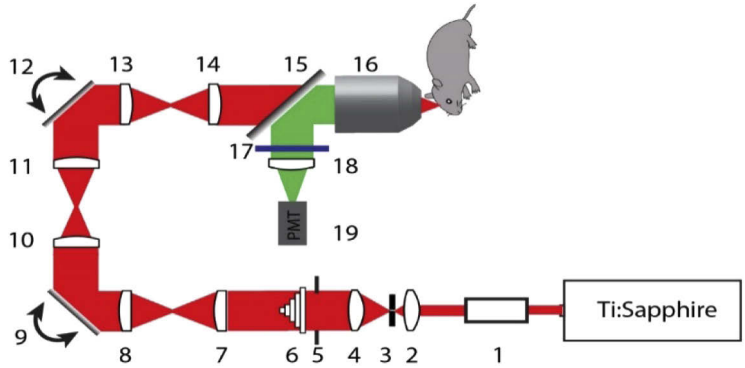

Fig. 4. Schematic diagram of the experimental setup. 1, Pockel cell, 350-80-2, Conoptics; 2, Lens, AC254-50-B, Thorlabs; 3, Pinhole, PH-100, Newport; 4, Lens, AC254-300-B, Thorlabs; 5, Iris; 6, Beam splitter mask, A12802-35-040, Hamamatsu; 7, Lens, AC254-125-B, Thorlabs; 8, Lens, AC254-75-B, Thorlabs; 9 and 12, Galvo mirror, 6215H, Cambridge technology; 10 and 11, Plössl lens, F-75 mm; 13, Scan Lens, SL50-2P2, Thorlabs; 14, Tube Lens, TL200-2P2, Thorlabs; 15, Dichroic Mirror, FF735-Di02-50.8-D, Semrock; 16, Objective, XLPLN25XWMP2, Olympus; 17, Filters, FF02-694/sp-25, FF01-527/70-25, Semrock; 18, Lens, AC254-45-A, Thorlabs; 19, PMT, H7422-40, Hamamatsu.

with a telescope consisting of two Plössl lenses (F-75 mm, built from two AC508-150-C each, Thorlabs) onto the second galvo mirror (6215H, Cambridge technology). A scan lens (SL50-2P2, Thorlabs) and tube lens (TL200-2P2, Thorlabs) were used to conjugate the galvos and the beam

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splitter mask onto the pupil plane of the objective (XLPLN25XWMP2, NA 1.05 / 25X Olympus). The beam magnification was chosen such that an NA of 0.67 was used for excitation, i.e. the beam underfilled the pupil of the objective. For additional experiments, the relay telescope between the beam splitter mask was either removed (Appendix C) or modified for using a higher excitation NA of 0.9 (Appendix D).

Fluorescence light was collected through the same objective, reflected by a dichroic mirror (FF735-Di02-50.8-D, Semrock), filtered by a short-pass filter (FF02-694/sp-25, Semrock) and a bandpass filter (FF01-527/70-25, Semrock) and focused onto a PMT (H7422-40, Hamamatsu Inc). The signals from the PMT were amplified and filtered with an amplifier (DLPCA-200, Femto) and then read in with a DAQ card (PXIe-6341, X Series, National Instruments). The microscope was controlled using Scanimage (Vidrio Technologies, LLC, VA).

3. Results

3.1. Point spread function measurements

To measure the PSF of our two-photon EDF system, we acquired 3D stacks of 200 nm green fluorescent beads (Polysciences, PA). We compared the obtained EDF PSF to a diffraction limited PSF obtained without the EDF beam splitter mask using an NA of 0.67 for excitation and a PSF

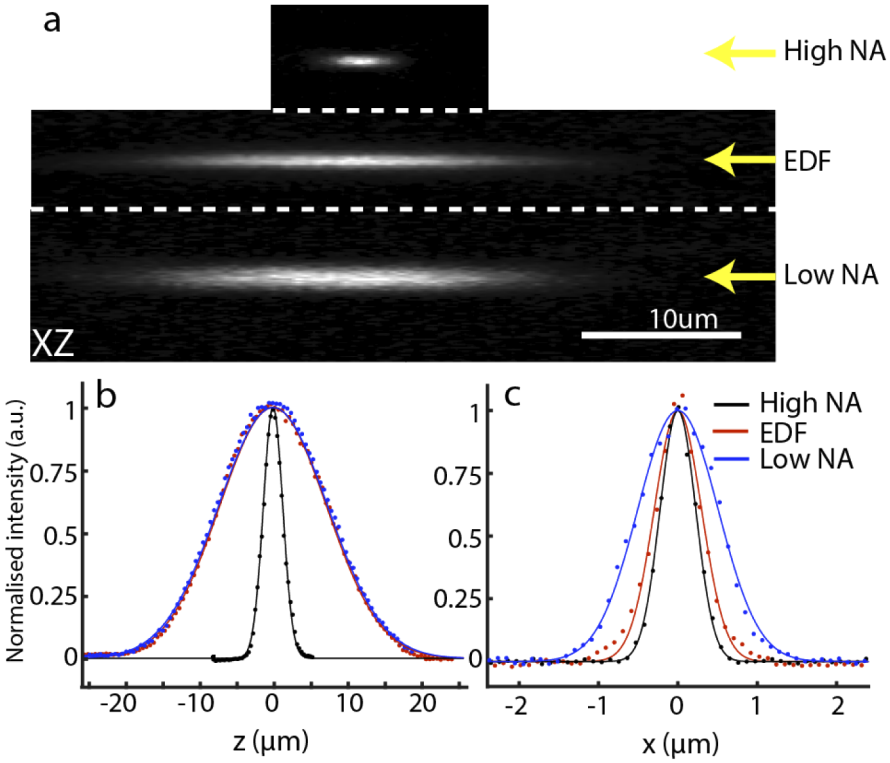

Fig. 5. Point spread functions measured with 200 nm green fluorescent beads. (a) Axial cross-sectional views of (top) high NA PSF for a laser focus using an NA of 0.67 and no beam splitter mask; (middle) PSF using an EDF beam splitter mask; (bottom) Low NA PSF corresponding to the innermost zone of the beam splitter mask. The low NA corresponded to 0.3 (b-c) Axial and lateral profiles through the point spread functions of high NA PSF without beam splitter mask (black), EDF PSF using the beam splitter mask (red) and low NA laser focus (blue).

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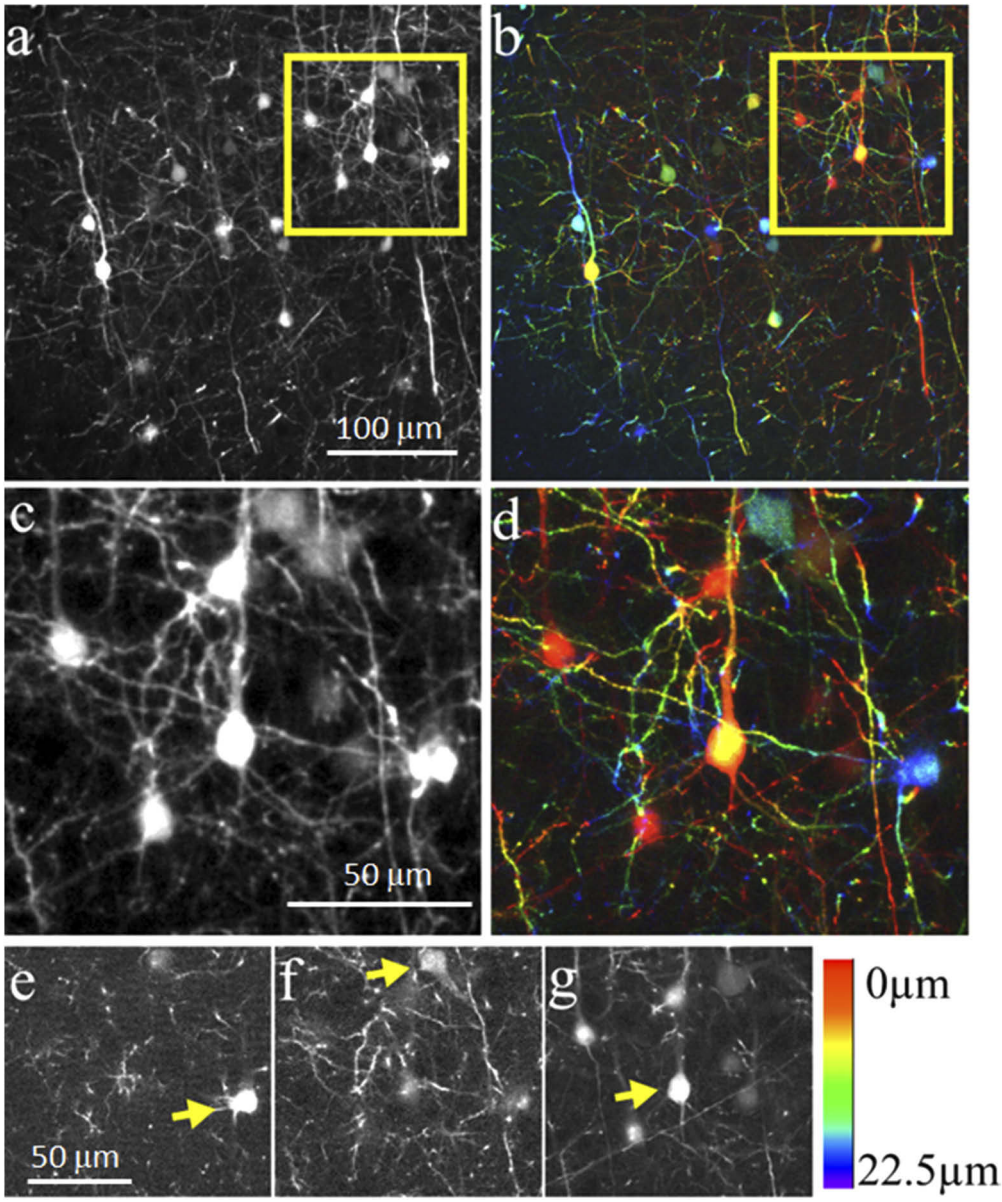

Fig. 6. Volumetric two-photon imaging of Thy1-GFP labelled neurons in fixed brain tissue using either the beam splitter mask or conventional two-photon laser scanning with z-stepping with an excitation NA of 0.67. (a) Single frame acquired with the beam splitter mask. (b) Projection of a 3D volume acquired with conventional laser scanning over 16 z-planes spanning a range of 22.5 microns. Color encodes z-position. (c) Magnified view of the boxed region of panel (a). (d) Magnified region of the boxed region of panel (b). (e-g) Individual z-planes from the region shown in panel (d) at 22.5 um, 10.5 um and 0 um depth. Arrows mark selected cell bodies that appear at these distinct depths.

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when the iris was closed such that only the innermost zone of the EDF mask contributed to the PSF, corresponding to an NA of 0.3 (Fig. 5(a)). The latter represents a low NA Gaussian laser focus, which had the same axial FWHM as the EDF PSF (Fig. 5(b)). The axial FWHM was 16.83 ± 0.29 µm (mean and standard deviation, n=5) for the EDF PSF and 2.90 ± 0.08 µm for the NA 0.67 excitation PSF without the beam splitter mask, which represented an axial extension by a factor 5.8. In the lateral beam profile, the EDF PSF was about 15% wider, increasing the width of the diffraction limited PSF of 0.53 ± 0.03 µm to 0.61 ± 0.02 µm (mean and standard deviation, n=5). By using a low NA beam to match the axial FWHM of the EDF PSF, the lateral width increased to 1.10 ± 0.06 µm (Fig. 5(c)).

3.2. Neuron imaging in fixed tissues

To demonstrate the potential of the beam splitter mask for neuroscience imaging applications, we imaged GFP labeled neurons in a fixed mouse brain tissue. Given the results of our simulations and PSF measurements, we set the excitation NA to 0.67. For fixation, a Thy1-GFP mouse was transcardially perfused first with cold PBS, then with 4% PFA in PBS as previously described [13]. The brain was isolated and further fixed in 4% PFA in PBS overnight at 4 °C. We first imaged a volume with the beam splitter mask as a single 2D image (Fig. 6(a)). Next, we removed the beam splitter mask and acquired a conventional 3D image stack encompassing a 22.5-micron deep volume using conventional z-stepping by moving the sample (Fig. 6(b)). Close examination of a selected region (Fig. 6(c)-(d)) revealed close correspondence between the lateral features in the EDF image and in the conventional 3D stack. The EDF image contained multiple cell bodies that were only present in specific z-planes (Fig. 6(e)-(g)) of a conventional acquisition. We noted that our conventional z-stack fitted best to the EDF data when using a range of 22.5 microns, which is larger than the axial FWHM of the EDF PSF (~16 microns). We explain this discrepancy by the assumption that in the EDF mode, some signal is generated outside of the FWHM range of the PSF. The laser power emerging from the objective was estimated to be 4.5 mW using the conventional laser focus and 21 mW for imaging with the EDF PSF.

4. Discussion

We have presented a simple way to extend the depth of focus in multiphoton microscopy, using a layered beam splitter mask previously developed for fluorescence widefield imaging. We showed that this mask design was advantageously used for extending the nonlinear focus in two-photon microscopy. The numerical and experimental results demonstrated that a five-fold extension in focal depth was feasible, without incurring sidelobes and with only a moderate lateral resolution loss. We noted that while the simulation predicted a lateral resolution loss in the order of 7%, in practice we measured a larger broadening of 15%. We also observed that with a larger excitation NA than 0.67, the discrepancy between simulation and experiment increased (Appendix C–D). We interpreted this as a sign that our optical train may have residual aberrations that did not manifest themselves as severely at a lower NA. Overall, we think that a medium NA of ~0.6-0.7 is best for applications of this EDF mask, offering a considerable increase in the depth of focus up to 16-20 microns while maintaining good optical performance.

By imaging fixed brain samples, we demonstrated that the method delivered faithful images that offered comparable image details as the projection of conventionally acquired 3D image data. As such, we anticipate wide applications of the beam splitter mask for rapid volumetric imaging where multiple objects at different depths must be monitored simultaneously (e.g. calcium or voltage sensing in neurons). Also, the increased depth of focus could be helpful in scenarios of strong sample motion, as a much larger z-range remains within focus.

The method works on the principle that the time delay between the individual zones exceeds the pulse duration of the ultrafast laser pulse. As such, the method is broadly applicable to a wide range of pulsed laser sources as long as interference between pulses can be avoided. As the

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exact phase shift between each beamlet is not of importance, it is expected that this method is insensitive to changes in the pulse wavelength. This is in stark contrast to methods that rely on phase masks that are precisely tailored to precise phase shifts for a specific wavelength.

Compared to two-photon raster scanning with Bessel beams, our method promises much higher light-throughput and reduced complexity. In fact, the mask can be directly placed in front of the galvanometric mirror (Appendix C). As such, an existing two-photon raster scanning microscope can be easily retrofitted without adding any other optical components than the mask. In contrast, published Bessel beam raster scanning designs rely on spatial light-modulators (SLM) or axicons, and annular masks, which require additional optics to conjugate these components properly. However, we note that two-photon Bessel beams with higher aspect ratios have been produced than what we achieved with a layered beam splitter mask as presented here. In principle, a mask with more annular zones could be produced to yield a larger gain in focus extension, which however would increase the complexity of the device. A main difference between the two types of beams is in our opinion that a two-photon Bessel beam can produce pronounced sidelobes [1], whereas the EDF beam presented here has a smooth, non-oscillatory transition into the background.

Bessel beams have also been hailed as being more robust to aberrations and optical occlusions. We do not know if some of these properties are also inherited for our EDF beam design. Since the effective focus is the superposition of many foci, which in turn each are essentially Bessel beams, it can be stipulated that such an arrangement is more robust to aberrations than interfering the whole wavefront coherently. We see another advantage for our method over Bessel beams in terms of adaptive optics. As our method uses the full pupil of the objective (in contrast, a Bessel beam only occupies a thin annulus), our method is expected to be readily compatible with adaptive optics that commonly use deformable mirrors conjugated to the pupil plane.

Lastly, it has been demonstrated that splitting an ultrafast laser beam into multiple, time-delayed beams can reduce the effects of photo-bleaching [14]. In this regard, our method represents a simple way to multiplex the laser pulse into five individual beams that arrive at slightly different times. As such, it would be expected that this could reduce photo-bleaching by a similar effect as previously published.

In conclusion, we present an unexpected use of a commercially available beam splitter mask for ultrafast optics that was originally intended for linear optics using low coherence fluorescence light. We find that for nonlinear techniques, including two-photon raster scanning fluorescence microscopy, the method has some advantages over other EDF approaches, such as high light-throughput, lack of sidelobes and being achromatic. We also think that due to its simplicity, this method will rapidly find widespread applications in multiphoton microscopy, as existing instruments can be readily retrofitted for EDF imaging.

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Appendix

A. Sidelobe comparison

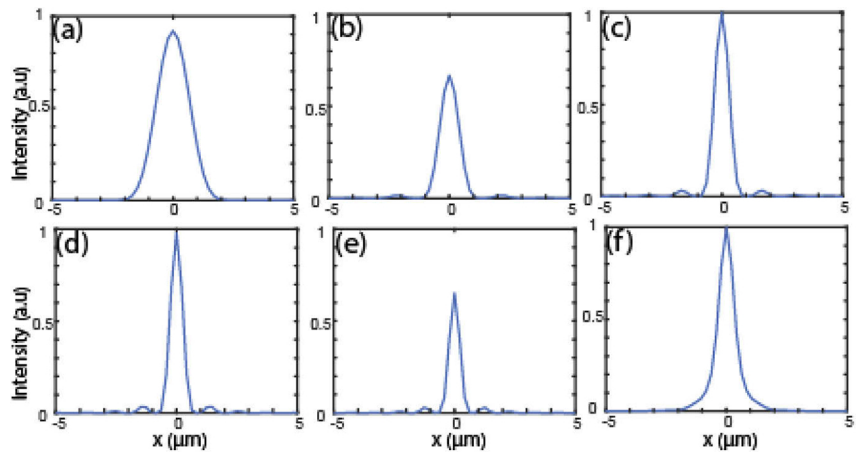

Fig. 7. Cross-sectional profiles of the PSFs shown in Fig. 1(c), centered at the waist of each beam. (a-e) Lateral profiles of PSFs arising from zone 1 through 5 of the beam splitter mask. (f) Corresponding profile of the EDF PSF. The intensities in (a-e) were normalized to the highest peak intensity of these five beams (profile shown in c). The EDF PSF was normalized separately to its own peak intensity.

B. Proof for the linear relation between the number of zones and focus extension

For the iterative procedure of determining mask zone sizes, we require that all zones have the same depth of field. This depth of field, d, for a wavelength $\boldsymbol{\lambda}$ is given by

$$d = \frac{\lambda}{n'(\cos\alpha_{\text{inner}} - \cos\alpha_{\text{outer}})}$$

where n' is the refractive index of the sample space, and α_{inner}, α_{outer} are the inner and outer zone angles, respectively. We require that the phase mask is continuous, such that α_{outer} of the innermost zone is α_{inner} of the next zone, etc. Hence, for N zones having $N+1$ different α_n, our depth of field requirement is such that

$$d = \frac{\lambda}{n'(\cos\alpha_{n-1} - \cos\alpha_n)}$$

This leads to the recurrence relation

$$\cos\alpha_n = \cos\alpha_{n-1} - \frac{\lambda}{n'd}$$

Recognizing this as an arithmetic sequence, we can use the general relation for the Nth term,

$$Z_N = Z_0 + N\delta$$

and, identifying that $\delta = -\lambda/n'd$ and $Z_N = \cos\alpha_n$, we have

$$\cos\alpha_N = \cos\alpha_0 - \frac{N\lambda}{n'd}$$

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As $\alpha_0 = 0$,

$$\cos\alpha_N = 1 - \frac{N\lambda}{n'd}$$

Rearranging for d, we find

$$d = \frac{N\lambda}{n'(1-\cos\alpha_N)}$$

This is exactly N times the depth of field obtained if the mask was simply one zone with the full size of the pupil, which is given by:

$$d^* = \frac{\lambda}{n'(1-\cos\alpha_N)}$$

Hence, we conclude that a pupil mask spanning the full size of the pupil and featuring N zones extend the depth of field of the objective N-fold. The same argument of course holds if one starts out with a truncated pupil, as it was done in this manuscript.

C. Direct placement of the beam splitter mask in front of the galvanometric mirror

A considerable simplification for retrofitting an existing two photon raster scanning microscope can be achieved by simply placing the beam splitter mask right next to the first galvanometric mirror. As such, the mask and this particular galvo cannot be both perfectly conjugated to the pupil plane. However, we experimentally could not see much difference in the PSF when we slightly moved our beam splitter mask axially, so there was some robustness for axial misalignment of the mask. We thus removed the telescope that relays the beam splitter mask to the first galvo

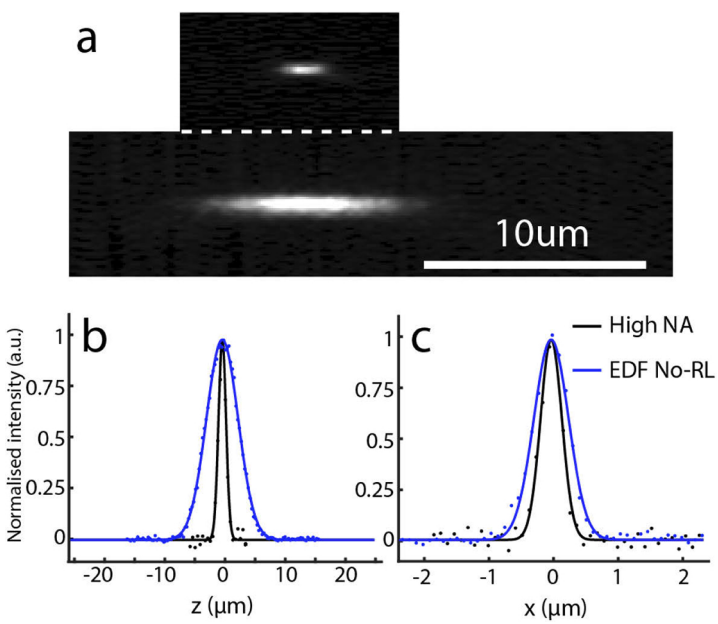

Fig. 8. Point spread functions measured with 200nm green fluorescent beads. (a) Axial cross-sectional views from top to bottom: PSF for laser focus using an NA of 1.05 and no beam splitter mask; EDF PSF using the beam splitter mask placed directly in front of the galvo mirror. (b-c) Axial and lateral profiles through the point spread functions. Black curves correspond to the high NA PSF without EDF mask and the blue curves correspond to the PSF using the beam splitter mask with no relay telescope (No-RL).

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mirror and placed the mask as close as possible to the galvo. By removing the telescope, we no longer underfilled the objective pupil, i.e. the full NA of 1.05 was used for excitation. Since the mask was larger than the diameter of the clear aperture of our galvanometric mirrors, only ~four zones of the mask could be used. The FWHM in the axial direction was 1.49 ± 0.03 µm when using the full NA of 1.05 and 6.02 ± 0.09 µm for the EDF PSF (see also Fig. 8) using the same NA for its outermost zone that could be used. The lateral FWHM increased from 0.38 ± 0.02 µm (NA of 1.05) to 0.52 ± 0.03µm for the EDF PSF (mean and standard deviation, n=5). We could observe the expected ~four-fold depth of focus increase for the EDF PSF. As such, we expect that a suitably sized beam splitter mask, which fits the clear aperture of the galvo mirror, can be directly used without an additional relay optics system.

In this configuration, the entire NA of the objective was used. To use an effective smaller excitation NA as presented in Fig. 5, two approaches would be feasible: Either a smaller beam splitter mask would need to be used to match the existing microscope, or larger galvo mirrors, together with a smaller magnification to the pupil plane, could be employed.

D. Point spread function with an upper NA of 0.9

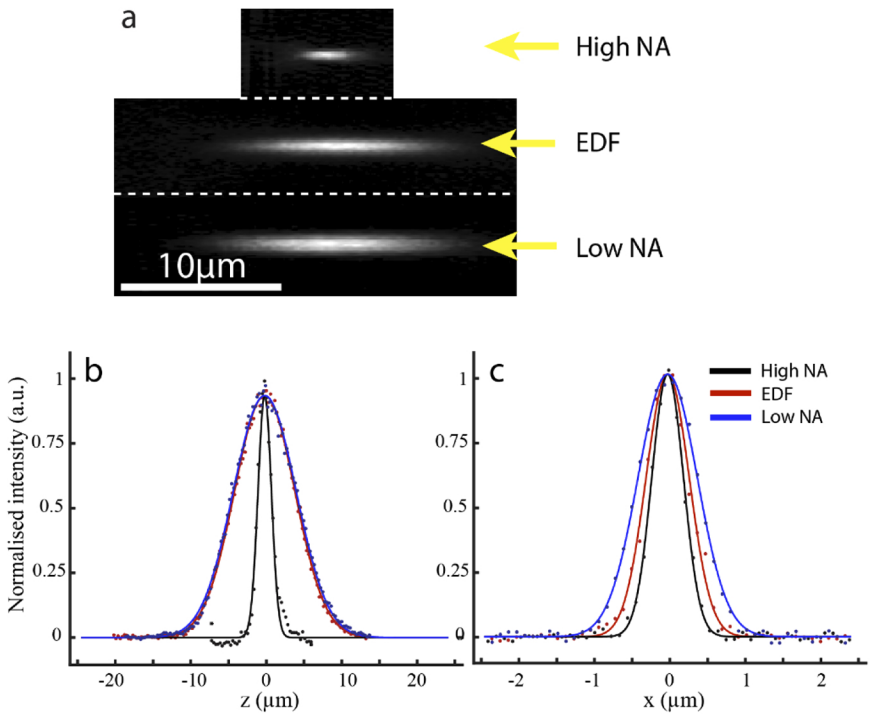

Fig. 9. Point spread functions measured with 200nm green fluorescent beads. (a) Axial cross-sectional views; (top) a laser focus using an NA of 0.9 and no beam splitter mask; (middle) EDF PSF using the beam splitter mask; (bottom) Low NA PSF corresponding to the innermost zone of the beam splitter mask. The NA corresponded to 0.4. (b-c) Axial and lateral profiles through the point spread functions of the high NA PSF (black), the EDF PSF using the beam splitter mask (red) and the low NA PSF (blue).

While we had our rationale why to use an excitation NA of 0.67, we further investigated how the beam splitter mask would perform when a higher excitation NA was used. To this end, the telescope relaying the beam splitter mask to the galvo was changed to lenses with focal lengths of 125 mm and 100 mm, which resulted in an effective excitation NA of ~0.9. We measured the

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resulting PSF with and without the beam splitter mask and the corresponding low NA Gaussian beam that matches the axial length of the EDF PSF (corresponding to an NA of 0.4) (Fig. 9(a)). The Full Width Half Maximum (FWHM) in the axial direction was 2.06 ± 0.13 µm (mean and standard deviation, n=5) for the high NA PSF with an NA of 0.9 and 10.17 ± 0.39 µm for the EDF PSF (Fig. 9(b)). The lateral FWHM increased form 0.42 ± 0.03 µm (high NA PSF) to 0.53 ± 0.03 µm for the EDF PSF. The low NA PSF had a lateral FWHM of 0.90 ± 0.04 µm (Fig. 9(c)).

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Disclosures

The authors declare no conflicts of interest.

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